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The Journal of Sports Medicine and Physical Fitness 2024 Sep 17
DOI: 10.23736/S0022-4707.24.16222-6

REVIEW

BODY COMPOSITION, NUTRITION AND SUPPLEMENTATION

Nutritional approaches applied to recovery of skeletal muscle injury immobilization: a review of nutrition aid for sport trauma

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ABSTRACT

Skeletal muscle (SM) injuries are a very common occurrence among athletes during training and/or competitive periods, resulting in time loss and absence from sports. Of several existing interventions to improve SM recovery, the nutritional approach has emerged as integral part to improve the physiological response and maintain the body composition to promote a rapid and safe return to play. Growing evidence supports that the physiological response to SM injuries results in a number of changes, suggesting the relevance of a personalized nutritional approach, according to the degree of immobilization, SM mass loss, strength and function. Therefore, we aimed to review the current evidence of the nutritional approach applied to SM injuries recovery in athletes, highlighting energy intake, use the main nutrients and elements for the SM recovery, such as proteins, carbohydrates, vitamins and omega-3 fatty acids. Currently, evidence from injured athletes is scarce, and so specific guidelines cannot be established, calling from greater research into nutritional interventions in SM injury recovery in athletes.

(*Cite this article as:* Rocha-Rodrigues S, Leão C, Marinho M, Afonso J. Nutritional approaches applied to recovery of skeletal muscle injury immobilization: a review of nutrition aid for sport trauma. J Sports Med Phys Fitness 2024 Sep 17. DOI: 10.23736/S0022-4707.24.16222-6)

KEY WORDS: Nutritional support; Athletic injuries; Wound healing; Muscular atrophy.

Sports-related injuries are very common affecting individuals of all ages, from amateur to large events, such as World Championships and/or Olympic Games. From all sports injuries, skeletal muscle (SM) injuries are among the most common,[1] accounting for 10% to 55% of all occurrences.[2, 3] The incidence of SM injuries is high, with the highest incidence in the lower limbs (*i.e.*, hamstrings, adductors, quadriceps and calf muscles).[4] Recent data shows SM injuries incidence rates ranging from 7% to 65% in soccer (upper limbs [10%], lower limbs [65%], body trunk [7%]), basketball (upper extremity [18.5%], lower extremity [58.7%], body trunk [8.4%]) and volleyball (upper extremity [30.5%], lower extremity [46.3%], body trunk (12.2%]).[5] If categorized according to muscle groups, hamstrings injuries are the most frequent (incidence of 17%) and are more likely (relative risk 1.4) in male than in female athletes.[6] The anatomical hamstrings characteristics, *i.e.*, biarticular organization, dual innervations of the biceps femoris and their muscle fiber-type distribution, negatively influences running coordination patterns, increasing the injury risk during running actions and/or eccentric SM contractions.[7]

The competitive and economic implications arising from SM injuries are of the upmost relevance for the

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athlete's health and performance as they result in a time loss and absence from sports, including both training and competing.[8, 9] Among soccer professional players, 27% of SM injuries contribute for time loss and absence (*e.g.*, 223 days of training absence and missing 37 matches).[4] Hereupon, SM injuries are a major concern in sports. The recovery from SM injuries is highly variable depending on several risk factors (*e.g.*, anatomical, physiological, biochemical, hormonal and biopsychological) being challenging to determine when athlete is ready to return to play.[9, 10] Moreover, a precocious return to play may be a critical aspect in the high re-injury rates and prolonged time loss.[4]

While research on medical, physiotherapy, and exercise-related practices are paramount for SM injury recovery, perhaps a "food first approach" can help to further enhance the recovery from SM injuries and also athletes' health and performance.[11] Growing evidence shows that the body's physiological response to SM injury leads to a change in the metabolic profile and energy needs,[12, 13] suggesting the need for a personalized nutritional approach, according to the degree of immobilization, physical activity levels, and the degree of SM mass loss, strength and function. During the SM recovery process (*i.e.*, inflammatory, regeneration, remodeling and repair phases), an adequate energy and dietary intake, including high-quality protein, antioxidants, omega-3 fatty acids, and probiotics should be ensured.[14] A special attention has been provided to the dietary protein due to its role in protecting SM mass and strength loss by controlling catabolism, antioxidants and anti-inflammatory compounds mediating the inflammatory cytokines secreted after injury.[14-17] Therefore, we aimed to review the current evidence of the nutritional approach applied to SM injuries recovery in athletes, highlighting energy intake and the use of main nutrients and elements for the SM recovery, such as proteins, carbohydrates, vitamins, and omega-3 fatty acids.

Skeletal muscle injuries

Classification

Clear grading systems for SM injuries classification are important for clinicians, coaches and athletes and should be used as prognostic and therapeutic purpose.[2] However, the literature proposes several different definitions and classification schemes for SM injuries.[1, 2, 18, 19] The heterogeneity and controversy[20] make difficult the comparison across studies that used distinct definitions and categorizations. In 1966, the American Medical Association published a simple clinical system based on three different grades for SM injuries as a mean for determining injury severity, classified as minor, moderate or complete injury (or grade I, II and III) through imaging evaluation and/or clinical aspects.[21] Although a lack in diagnosis accuracy and the prognostic information provided for the clinician (*e.g.*, length of muscle tear on magnetic resonance imaging (MRI), MRI 'negative' injuries, distance from the origin, cross-sectional area of edema and involvement of the tendon),[2, 20] this simple grading system remained popular and until recently largely used among clinicians.

Recently, the Munich consensus[19] and others[1, 18] have proposed alternative SM injuries grading systems. Briefly, the system proposed by Pollock *et al.*[18] (British athletics SM injuries classification [BAMIC]) uses the anatomical location and extension of the lesion through imaging. Injuries are graded 0-4 based on MRI features, with grades 1-4 including an additional suffix "a," "b," or "c" if the injury is "myofascial," "musculo-tendinous," or "intratendinous." The classification proposed by Valle *et al.*[1] seek out for 4 characteristics of SM injury based on "MLG-R" that each letter means: mechanism of injury (M), location (L), degree of injury (G) and number of re-injuries (R). In addition, both systems consider clinical aspects, such as intensity, time of onset, and location of pain, to define the type of lesion and provide an adequate prognosis.

The classification proposed by Mueller-Wohlfarht *et al.*,[19] also known as the Munich Consensus, consider etiological aspects. This one classifies SM injuries as: 1) direct, caused by contusion or laceration; or 2) indirect, subclassified into functional (nonstructural) or structural. Functional subtype is related to fatigue induced, delayed onset muscle soreness (DOMS), spine-related neuromuscular dysfunction or SM-related neuromuscular dysfunction. Whereas structural subtype is related to muscle pathology, which is divided into three essential grades of SM injury: minor, moderate and complete. However, this system has some limitations as clinical features of this pathology are categorized as structural rather than functional, and important "functional" elements to the diagnosis should be considered.

As each new classification and grading systems emerged, a little consensus on how to either effectively describe or determine the prognosis of a specific SM injuries exists. So far, studies published did not allow to define a graduation of SM injury since mechanisms of injury and recovery time vary for each SM group. Thus, these misunderstanding difficulties the creation of a homogeneous system with a good correlation with clinical outcomes.

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Skeletal muscle recovery

Following acute SM injuries, fibers retract and small blood vessels tear, filling the injured area with blood, and creating a clot known as hematoma.[22] The resident stem cells, also known as satellite cells, are primarily responsible for muscle regeneration along with a well-regulated inflammatory cascade.[22, 23] The dynamic interaction between SM tissue and the immune system is directly responsible for the proper regeneration following soft tissue trauma.[24] Such injuries with local inflammation, healing process typically follows: 1) an inflammatory phase (or destruction phase, 0-7 days post-injury); 2) a regeneration (4-14 days post-injury); and 3) a remodeling and repair phase (14-28 days post-injury).[24]

During the inflammatory phase, hematoma serves as a support to potentiate healing, ensuring that the repair occurs only in the injured site. Afterwards, a coordinated activity between inflammatory mediator-like cytokines and growth factors (*e.g.*, growth-factor transforming, platelet-derived growth, hepatocyte growth factor, heat shock protein 70, vascular endothelial growth factor, fibroblast growth factor), satellite cells and immune system cells (*e.g.*, neutrophils, monocytes, and macrophages) into the extracellular matrix.[25] Depending on the degree of injury severity, a decreased physical activity or immobilization may be recommended for minimizing pain, decreasing the effects of the destruction phase to the site of injury, speed of the recovery and fosters a most complete recovery.[24, 26]

The SM regeneration is started immediately following the hematoma,[22] from 4 to 5 days post-injury until 14 days post-injury. In this phase, an enriched pro-inflammatory environment in the injured site contributes to regulating satellite cells functions and formation of new capillaries (*via* endothelial nitric oxide synthase), increasing blood flow to the site of injury occur.[24]

Both remodeling and repair phases occur concomitantly with the SM regeneration phase. Here, the regenerating SM fibers and connective tissue maintained into mature state and a preliminary fibrous scar of wound healing is formed.[24] The scar serves as a template for collagen deposition and as a platform for cell migration and cell growth.[23] Typically, SM tissue is oriented in straight lines, so, a treatment during this phase can assist the new tissue to regenerate into parallel lines.[24] Potentially, these different recovery phases may benefit from distinct nutritional approaches, which will be explored later in this review.

The acute management of SM injuries recovery were guided by acronymous, such as ICE, RICE (rest, ice, compression, and elevation), PRICE (protection, rest, ice, compression, and elevation) or POLICE (protection, optimal loading, ice compression, and elevation),[27, 28] but their evidence on sub-acute and chronic stages of tissue healing is limited. Recently, a new acronymous arose encompassing the recovery continuum from immediate stage (PEACE) to subsequent management (LOVE). The PEACE&LOVE approach underlines on the education of patients, psychosocial factors, and avoids anti-inflammatories medications.[29] Discussing these approaches, as well as the specific medical and exercise-based interventions during SM injury, is beyond the scope of this review. Instead, we will turn out attention to potential nutritional strategies applied as adjunct to enhancing SM injury recovery.

Nutritional strategies after muscle injury

One of the pillars of an athlete's health and performance is their nutrition. The primary purpose of an athlete's diet is to achieve adequate energy and macronutrient/micronutrient intake to facilitate the optimal restoration of their functional and athletic capacities.[30] Nutrition has been described as an integral part of the recovery for the exercise-induced SM damage,[10] which may occur after exertion (*e.g.*, after a training session). Theoretically, an adequate nutrition may enhance muscle recovery, prevent or reduce SM mass loss, reduce inflammation, and overall provide nutrients for optimal athlete's recovery.[13, 31-33] Indeed, evidence shows that the body's physiological response to SM injury leads to a change in the metabolic profile and energy needs,[12, 13] requiring a personalized nutritional intake, according to the degree of immobilization, physical activity levels, and the degree of SM mass loss, strength and function.[14] However, there is limited research for prevention and treatment of SM injuries in athletes, despite suggestions that an "optimal" nutritional approach can help in all the phases of recovery.[10] The knowledge of how nutritional approaches may affect recovery times and accelerate return to sports is crucial.[24]

Energy intake

Both regeneration and remodeling phases are determinant for energy intake due to hypermetabolism response to SM injury.[23] An increased blood pressure, heart rate, peripheral insulin resistance, protein and lipid catabolism can lead to increased resting energy expenditure, body temperature, total body protein loss, muscle wasting, and stimulated synthesis of acute-phase proteins.[25] The energy expenditure is more dependent on the degree of the inflammatory process than the extent of the injured tissue.[34] Due to the

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healing process, energy expenditure might represent up to 50% of the basal metabolic rate or in the case of crutches utilization that can boost the deambulation energy cost,[23] suggesting that hypermetabolic response to injury has a significant impact on energy expenditure. An increased between 15% to 50% of energy expenditure was estimated to occur in all trauma, surgically or critically ill patients,[35] but the magnitude of this response is particularly severe and sustained in burn injury.[36] Although there is no evidence on the increase in energy needs in injured athletes, it has been suggested that this increase may occurs due to an augmented basal metabolism, specifically due to the growth and repair of tissues.[37]

The importance of total daily energy expenditure is reflected by the consequences of energy imbalances, either excess or deficit. Two studies conducted by Biolo *et al*.[12, 13] evaluated the impact of hyper and hypocaloric diets on healthy individuals under different conditions: activity, ambulatory, and bed rest. In the study,[12] resting energy expenditure relative to lean body mass did not differ significantly after 14 weeks of controlled ambulatory and bed rest (6° head-down tilt) conditions in sedentary male individuals (N.=9; 24±1 years; 77±2 kg and Body Mass Index of 23±1 kg/m^2) receiving eucaloric or hypocaloric (~80% of total energy expenditure) diets. In the study,[13] 19 healthy male individuals (24±1 years and Body Mass Index of 23.5±0.6) were tested before and after 5 weeks of bed rest. During this period, individuals received a decreased energy requirement (study A, N.=10) or activity-marched diet (study B, N.=9) diets, according with higher or lower energy balance obtained from inactivity-induced changes in fat mass. As expected, hypocaloric diet regimen induced a great wasting of lean body mass in bed rest conditions and a higher leucine deposition rate within the total body protein (protein synthesis) in ambulatory condition.[13] Moreover, the hypercaloric energy balance and muscle atrophy have been associated with the activation of systemic inflammatory response and antioxidant defense under stress conditions.[12] Individuals under hypercaloric diets exhibited a higher body fat mass and SM mass loss.[12]

These data highlight the potential critical role of consuming appropriate amount of energy and the levels of physical activity during the recovery of SM injuries since both have functions in physiological response (*i.e*., increased inflammatory processes, repair of injured tissue and remodeling/healing) as well as in maintenance of SM mass and function.[12, 13] However, we highlight that these data were obtained from healthy individuals and hence should not be generalized to injured athletes, so further studies are required to clear the role of metabolism during SM healing process.

Moreover, the energy expenditure during SM injury recovery is highly variable, depending on the type, degree and extension of SM injury, additional structures affected (e.g., ligaments, aponeurosis, neurovascular structures), degree of immobilization and the type of support that the athlete will have access to during the recovery process.[30] So far, the recommendations for daily energy needs for an optimal energy expenditure and SM recovery, without accumulating an excess of energy that can change the body composition (*e.g*., increased body fat) are unclear. In this line, instead of estimating the energy expenditure of an injured athlete, it is recommended to monitor the body fat percentage to get more reliable information about energy balance than using less reliable estimation methods, and a personalized nutritional approach according to individual's age, sex, weight and physical condition are also recommended.[12, 38]

Macronutrient intake

Protein

The main concerns in SM injuries and their consequent immobilization or limited mobility/physical activity encompass reduced SM protein synthesis, anabolic resistance, proteolysis and loss of strength.[39] In healthy men under energy balance conditions, one week of bed rest reduced muscle mass by decreasing 1.4±0.2 kg of lean tissue loss, 3.2±0.9% of quadriceps cross-sectional area, and 29±5% of whole-body insulin sensitivity.[40] After two weeks of immobilization, *in vivo* rate of SM proteins synthesis was impaired, and an anabolic resistance to protein ingestion contributed to the loss of muscle mass, in recreationally active young men (N.=12; 24±1 years).[39] The most existing evidence was obtained from studies conducted in immobilized patients and/or patients with limited functional activities,[41] old individuals[42] studies, or in a case study of an injured athlete submitted to immobilization.[43]

During SM injuries, nutritional approaches during immobilization and recovery periods could focus on: 1) providing anabolic factors and improve amino acids availability (*i.e*., nutrients that can achieve stimulation of protein synthesis); and 2) decreasing the threshold of anabolic resistance through restore the muscle sensitivity to the stimulant effect of food intake, especially in the postprandial phase.[39, 44-46] The current International Society of Sport Nutrition position stand recommend an ingestion of

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a protein dose of 20-40 g (0.25-0.40 g/kg body mass/dose), including in all meals a high-quality source every 3 to 4 h during the day.[47] Also, the ingestion of 10 g/day of essential amino acids in free form or as part of the meal timing above maximizes the stimuli of muscle protein synthesis.[47] In a randomized double-blind, placebo-controlled trial, supplementation with leucine (0.06 g/kg/meal, in total of 0.18 g/kg/day) preserved partly muscle health of middle-aged during 14 days.[46]

Again, though, while large-scale studies with injured athletes are not performed (preferably randomized), we should refrain from any strong statements concerning the role of nutrition, and particularly protein intake in SM injury recovery.

Carbohydrates

The availability of carbohydrates has an impact on protein synthesis and SM mass and strength in response to exercise;[48] however, its role during recovery from a SM injury is little known. Extensive literature supports a protective effect against protein catabolism by reducing protein breakdown, whereas a high-fat diet likely not induce the same protective effect.[11] In the study,[49] a randomization was performed to stratified 26 elite male runners into three different groups, as follows: low (60 g/h of carbohydrates, N.=8; 37.8±9.4 years; 71.8±10.3 kg), control (90 g/h of carbohydrates, N.=9; 37.2±5.4 years; 66.6±10.8 kg) and experimental (120 g/h of carbohydrates, N.=9; 38.0±6.8 years; 67.4±11.1 kg). They reported that high amount of carbohydrates (120 g/h) *versus* amount recommended (90 g/h) limits the increase of exercise-induced muscle damage markers, such as creatine kinase, lactate dehydrogenase, and glutamic oxaloacetic transaminase, during a mountain marathon.[49] Although this study did not observe those alterations from a SM injury, it is interesting to verify that high doses of carbohydrates (vs. recommendations) had beneficial effects against exercise-induced muscle damage.

The most interesting evidence regarding carbohydrate consumption in injured athletes derives from two case studies comparing different nutritional options in soccer players' during recovery phase from anterior cruciate ligament injury.[37, 43] In the Milsom *et al*. study,[43] between the 8th and 24th week inactivity and immobilization, the athlete (soccer player) adhered to a moderate carbohydrate-high protein diet combined with structured resistance and field-based training for both the lower and upper-body that resulted in whole-body muscle hypertrophy (varying from 0.5 to 1 kg per week). Both studies[37, 43] suggest that it could be valuable to avoid excessive reductions in energy intake during the initial weeks post anterior cruciate ligament (ACL) surgery, to limit muscle atrophy. These studies[12, 13] were both conducted in soccer, both in the Premier League, and both reported ACL injuries. As such, findings should be looked at carefully and cannot be generalized to other injuries or sports. Therefore, the role of carbohydrates in the SM injury recovery remains scarcely researched.

Lipids

Lipids are an important source of energy that are involved in a plethora of biological processes in the body, assuming numerous functions, from the functioning of cell membranes, gene activation, modulation of signalling pathways and for elevating cell proliferation,[50] and also in fat soluble vitamins' absorption.[51] Evidence suggests that a recommended proportion of 20-25% of total caloric intake should be derived from dietary lipids: 0.8 to 2 g/kg/day.[52] Lipids have a higher energy density than carbohydrates.[53] Thus, lipids intake is important for maintaining energy balance, but a high consumption can lead to undesirable fat mass increase while a lipids restriction may compromise some processes in which it participates, namely hormone production and inflammation modulation.[31, 53] A position statement from American College of Sports Medicine for nutrition and athletic performance recommends not to restrict lipids intake below 20% of the total caloric intake.[31]

Omega-3 fatty acids are the most studied type of lipids due to their potential role in inflammation.[54] However, the inflammatory response at an early stage have been described as crucial for SM healing, which may suggest that the omega-3 supplementation may compromise the physiological evolution of injury recovery.[53] On the other hand, an augmented inflammation at some degree may also difficult the recovery process.[55] Eicosapentaenoic acid (EPA) and docosahexaenoic acid are the most commonly omega-3 polyunsaturated fatty acids with an interesting role in reducing inflammation via their incorporation into membrane phospholipids, where they block the production of pro-inflammatory eicosanoids, reducing of activation of immune cells and the associated release of pro-inflammatory cytokines.[56]

In this line, these features may also support some of their other beneficial effects in SM, such as altering protein metabolism, increasing muscle strength and modulating myogenesis. After an incubation with high concentrations (400-600 uM) of EPA for 24 h, an increased peroxisome proliferator-activated receptors, an important

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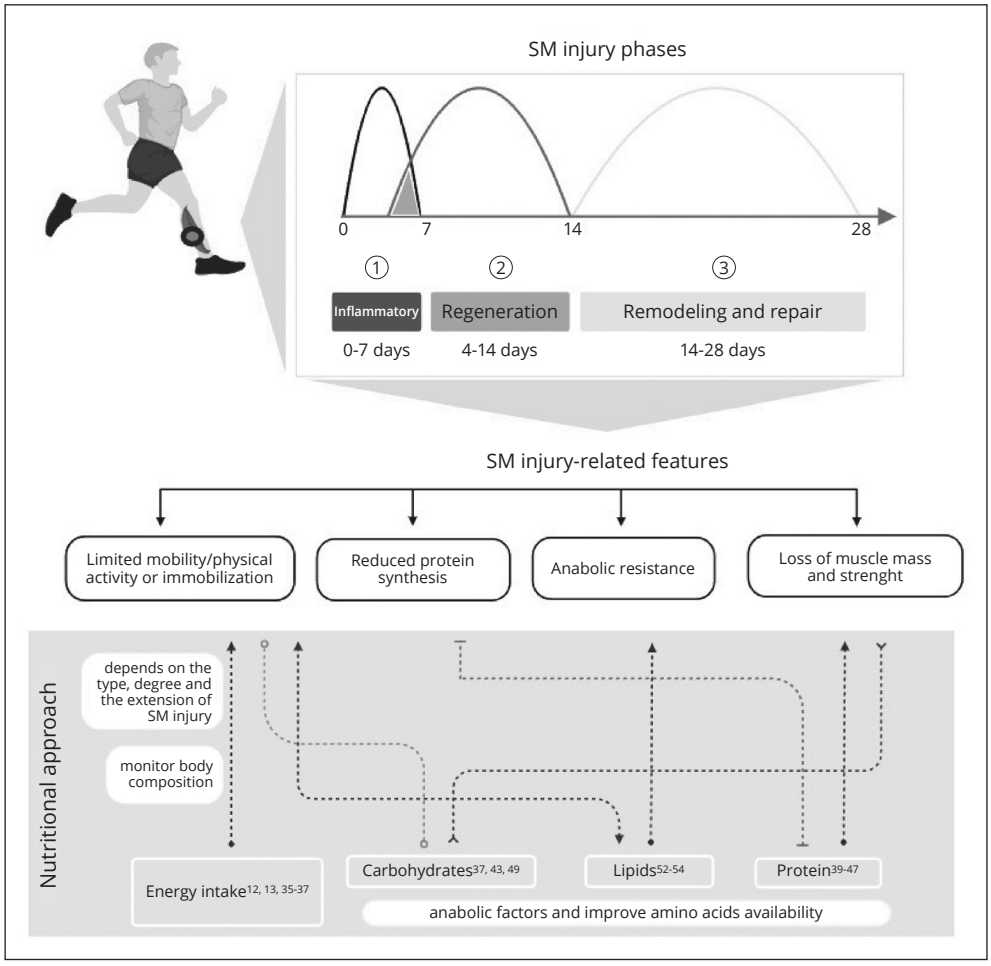

Figure 1.—Schematic view of current knowledge regarding nutritional approach on skeletal muscle (SM) recovery after an injury. Note that the most studies were conducted on laboratory-induced damage to study DOMS, being the knowledge of nutritional approach to SM injury limited.

transcriptional factor with roles in the regulation of cellular differentiation, development and metabolism, was observed in C2C12 myotubes.[57] These findings suggest that omega-3 could be a good candidate to consider in future research that should focus on the dose, timing and also the inflammatory responses from injury. Once again, however, it should be stressed that we currently have no evidence concerning the effects of lipids intake on athletes undergoing SM injury recovery (Figure 1).

Micronutrients

Minerals, trace elements and vitamins play a great role on protein, lipid and carbohydrates metabolism by activating several biochemical pathways and facilitating macronutrient absorption.[58] The recommended daily needs for these nutrients are defined for the general population;[59] however there is a gap how these recommendations should be apply to athletes. Unsurprisingly, the current knowledge regarding the importance of micronutrients in SM injury recovery is still scarce.[60]

Recently, much attention has been paid to vitamin D since the knowledge that SM is a direct target, mainly for its deficiency in general population and in athletes.[61] Observational studies have been suggested that low levels of 25-hydroxyvitamin D correlated with functional recovery of SM following eccentric contractions.[62] In a randomized placebo-controlled trial, supplementation with vitamin D (4000 IU/day) for 6 weeks contributed to the SM regeneration by enhancing reparative processes and potentially prevent SM mass loss, in physically active males (N.=20) with low levels of 25-hydroxyvitamin D.[62] Using the same study design, Zebrowska *et al.*[33] reported that 3 weeks of vitamin D supplementation (2000 IU/day) decreased post-exercise inflammatory and SM damage markers (*i.e.*, tumor necrosis factor and creatine kinase), in ultra-marathon runners (N.=24).

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Other vitamins, such as C and E, have also been studied in the SM damage due their role in offset muscle soreness during competitive periods.[63] Nevertheless, no evidence-based rationale exists to justify their use in an adaptive setting in athletes with normal vitamins levels. Additionally, a recent Cochrane review including 50 studies (N.=1089, male and female with range of age of 16-55 years) found that only moderate-to-low quality evidence supports the use of "antioxidant" supplements in reducing delayed onset muscle soreness.[63] It is believed that a reduction in inflammatory markers and the production of free radicals can lead to the perverse effect by interfering with physiological mechanisms related to recovery.[64] Using a double-blind and randomized study, Gomez-Cabrera et al[32] studied 14 sedentary male individuals (aged between 27 to 36 years) assigned into two groups: i) exercise performed aa static bicycle for 8 weeks (3d/week, 40 min at 65%-80% of $\dot{V}O_2$max) and ii) exercise supplemented with a daily dose of vitamin C (1g/day) taken at 9:00 a.m. The group of exercise plus vitamin C showed a decreased endurance training efficiency. Gomez-Cabrera et al 32 also studied the effects of vitamin C in exercised male Wistar rats (N.=24) submitted to 2 different protocols for 3 and 6 weeks. Data from rat skeletal muscle demonstrated a decrease of key transcription factors that are induced by exercise, such as mitochondrial biogenesis markers (e.g., peroxisome proliferator-activated receptor co-activator 1, nuclear respiratory factor 1, and mitochondrial transcription factor A) and antioxidant enzymes (superoxide dismutase and glutathione peroxidase),[32] suggesting that vitamin C supplementation decrease training efficiency due it prevents some cellular adaptations to exercise.

Overall, the pattern remains: beyond the recommendations provided for the general population, this is few additional evidence for the effects of micronutrients intake on SM injury recovery.

Supplements

Creatine is a compound with functions in cellular metabolism, being primarily used as a source of energy to resynthesize adenosine triphosphate.[65] As 95% of all creatine can be found in SM, with endogenous creatine synthesis responsible for providing about half of the daily need for creatine.[66] Historically, creatine has been proven to be a supplement with a significant impact on SM mass and strength gain as well as recovery after intense exercise, contributing to less SM damage in response to high-intense exercise. For these reasons, is the most recommended supplements for athlete,[67] but their intake during SM recovery in injured athletes is still unknown.

Studies focused on short periods immobilization in healthy individuals.[68-70] Using a single-blind crossover design, creatine supplementation (4x5 g/d) over 7 days maintained lean tissue mass, endurance strength vs. placebo group (ingested maltodextrin, 4x5 g/d), in young men (N.=7; 18-25 years) with upper-arm immobilized.[68] In young healthy individuals (N.=22; 13 males and nine females) with leg immobilized during 2 weeks, a double-blind trial was performed where half of the individuals received creatine monohydrate (from 20 g/d down to 5 g/d) whereas others ingested placebo (maltodextrin).[69] Findings suggest that oral creatine supplementation enhanced the recovery of SM mass by restoring SM mass after 2 weeks of immobilization and the anabolic effects of creatine were observed in both fast and slow twitch muscle fiber types.[69] Another study found no alterations induced by creatine supplementation 5 days (20 g/day) prior to short-term leg immobilization in SM mass or strength, in healthy young male individuals (N.=30; 23±1 years and Body Mass Index of 23.3±0.5 kg/m^2).[70]

Whey protein supplementation can be essential for the maintenance of SM strength after a long period of immobilization for their effects by attenuating the increase of physiological markers of SM damage, such as creatine kinase and lactate dehydrogenase, and, thus, contribute to the SM recovery.[71, 72] Young male professional soccer players (N.=24; 23±1 years and body index mass of 23.3±0.5) engaged in an official tournament treated with hydrolyzed whey protein (20 g/d) for 5 days exhibited significant reductions in SM damage indicators, such as creatine kinase (-42%) and lactate dehydrogenase (-30%), compared to maltodextrin intake, which increased muscle breakdown markers.[71] Other study addressed the effects of hydrolyzed whey protein (2 doses of 70 mL) or isoenergetic carbohydrate for 4 days post- exercise-induced SM damage, in physically active females (N.=20; 20±1 years and 61.8±7.9 kg).[73] In this study, a decrease in creatine kinase levels were greater following hydrolysed whey protein compared with carbohydrate at 48 h post-exercise-induced SM damage.[73]

What do we currently know about nutritional approach in SM injury recovery?

Currently, data regarding nutritional approach on SM recovery after an injury is very limited, with most studies conducted on laboratory-induced SM damage to study

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DOMS. Even thought that studies using this model provides a relevant insight about nutritional approaches, there are significant differences between DOMS and SM injuries at structural and functional levels.[71, 73] As opposite to SM injury, in a laboratory-induced SM damage model, individuals experienced soreness that usually increases in intensity during 24h after the activity, peaks from 24h to 72h, and then diminishes and disappear very soon. A SM injury typically implies local inflammation and follows 3 phases (inflammatory, regeneration and remodeling/repair) and the time of recovery generally lasts between 7 days to 21 days post-injury.[24]

Other studies were conducted in individuals submitted to very specific conditions, like immobilization or limited mobility/physical activity, which does not mimic SM injuries as some of them do not require immobilization, but partially limited mobility or reduced levels of physical activity. For instance, data from energy intake was mainly obtained from healthy individuals that were submitted to a short or long periods of immobilization or bed rest. Thus, these data should be not generalized to injured athletes.

Studies that focused on athletics' population, almost all included male, with data from female athletes being scarce. Although the female athletes have overall lower risk of SM injury, data from Dannecker *et al*.[74] reported a few, but large magnitude, sex differences after exercise-induced muscle pain and damage. Surprisingly, females described moderately lower and less frequent muscle pain than males. Thus, these findings point out the relevance of a personalized nutritional approach according to individual's age, sex, weight and physical condition that should be considered in future studies.

Conclusions

Current evidence shows that further clinical and experimental studies regarding nutritional approach and SM injury recovery are needed to better understand the effective role of energy and dietary intake in an injured athlete. Therefore, distinct nutritional strategies could be used as a part integrant along with personalized training and SM recovery programs, in athletes.

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Conflicts of interest
The authors certify that there is no conflict of interest with any financial organization regarding the material discussed in the manuscript.

Authors' contributions
Substantial contributions to the conception of the present review: Sílvia Rocha-Rodrigues, César Leão, Margarida Marinho, José Afonso. Draft the work or revising it critically for important intellectual content: all authors. All authors read and approved the final version of the manuscript.

Acknowledgements
The authors would like to thank to the individuals and organizations for their support, expertise, and encouragement that contributed to the overall quality and depth of our work.

History
Article first published online: September 17, 2024. - Manuscript accepted: July 16, 2024. - Manuscript revised: June 25, 2024. - Manuscript received: April 18, 2024.